An Adaptive Explainable Cognitive Graph Digital Twin for Cyber-Resilient Smart Hospital Energy Ecosystems

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Abstract. Smart hospital campuses rely on localized microgrids to sustain intensive-care ventilators, surgical robotics, and biomedical cold chains. A stealthy False Data Injection Attack (FDIA) on substation Remote Terminal Units can corrupt the energy-management state estimator while evading its χ^2 residual check, placing patient-critical loads at risk. We introduce a Cognitive Graph Digital Twin (CGDT) that mirrors the hospital electrical topology as a dynamic admittance graph and runs a multi-task Admittance-Conditioned GraphSAGE backbone for simultaneous attack detection and bus-level localization in a single forward pass at 0.908 ms latency. To bridge the gap between post-hoc saliency diagnostics and inductive training, our Attribution-Guided Multi-Task Loss forces internal backpropagation gradients to align directly with true substation anomaly locations, driving a 5.52 percentage-point improvement in localization F1. Dynamic physical-to-virtual coherence is governed by a state-space tracking filter coupled with an adaptive exponential baseline. Upon detecting an attack, a closed-loop healing module quarantines compromised sensor channels and reconstructs measurements via graph-guided spatial interpolation, reducing Weighted Least Squares estimation error by 99.97%. On the IEEE 14-bus test feeder under full nonlinear AC power flow, the CGDT achieves $98.25\% \pm 0.34\%$ detection F1 and $97.62\% \pm 0.84\%$ localization F1 across five random seeds, and retains 92.80% detection F1 under zero-shot topology reconfigurations without retraining.

Keywords: Graph Digital Twin · Healthcare Cybersecurity · GraphSAGE · Explainable AI · False Data Injection Attacks · Smart Hospital Microgrid

1 Introduction

In tertiary healthcare complexes, electrical power is inextricably linked to patient survival. Modern hospital campuses rely on localized microgrids that combine

utility interconnections, rooftop photovoltaics, battery storage banks, and diesel generators to energize critical medical loads [5,19]. These loads cannot tolerate instability: surgical robotics demand millisecond voltage consistency, intensive-care ventilators require uninterrupted feeds, and vaccine cold-chains depend on constant refrigeration at $-80°$C. Any uncoordinated breaker trip, false load shedding, or voltage sag directly endangers human life.

Substation Remote Terminal Units (RTUs) stream voltage magnitudes, phase angles, and power flows (V, θ, P, Q) into Weighted Least Squares (WLS) state estimators over IEC 61850 substation buses. Control centers routinely rely on normalized χ^2 residual tests to weed out random transducer noise and faulty metering. Yet this classical check fails entirely against coordinated vectors. By exploiting the topology-dependent measurement Jacobian H, an adversary synthesizes an injection $a = Hc$ that shifts state estimates by an arbitrary vector c while leaving algebraic measurement residuals entirely intact [15,13]. The corrupted telemetry slips past the detector unflagged; automated dispatch software then acts on compromised grid profiles.

Prior security schemes leaned heavily on tabular classifiers—ranging from support vector machines to multi-layer perceptrons and fixed-kernel convolutional nets [12,3]. These tabular baselines break whenever the grid flexes. Hospital distribution feeders undergo frequent branch switching, tie-line transfers, and transformer tap adjustments. Each physical breaker toggle changes sensor ordering, blinding fixed-input neural networks or triggering false tripping alarms. Retraining these models on the fly during 2–4 s SCADA sampling windows remains out of the question.

Graph neural networks (GNNs) solve this layout rigidity by treating substations as nodes and physical lines as edges [21]. Tripping a breaker simply toggles an entry in the graph adjacency matrix without disturbing node vector dimensions. While spatial wavelets [2] and isotropic message passing [22] accommodate switching events, conventional GNNs treat all incident branches identically. They ignore complex branch admittances ($Y_{uv} = G_{uv} + jB_{uv}$). By averaging nodal representations isotropically across neighbors, standard convolutions wash out high-frequency voltage and angle discrepancies across adjacent electrical corridors.

Clinical facilities tolerate neither unverified black-box alerts nor sluggish incident response. Perturbation-based explainers such as SHAP or LIME require hundreds of Monte Carlo passes, exceeding real-time SCADA deadlines by orders of magnitude [27]. Moreover, issuing passive alarms without mitigating the bad data leaves life-support circuits exposed to automated shedding. To safeguard critical healthcare microgrids, an anomaly framework must pin down compromised buses within sub-second intervals and synthesize clean replacement telemetry before downstream protection relays trip.

1.1 Key Contributions

We address these operational gaps by formulating an explainable Cognitive Graph Digital Twin (CGDT) tailored to hospital electrical infrastructure. Specific technical contributions include:

- **Physics-Informed GraphSAGE Formulation**: Rather than relying on unweighted adjacency matrices, we condition neighborhood message passing and spatial pooling on complex branch admittances ($Y_{uv} = G_{uv} + jB_{uv}$). This preserves sharp electrical gradients across low-impedance paths and prevents isotropic low-pass feature blurring.
- **Continuous Twin Synchronization**: By pairing steady-state Kalman gain tracking with an exponential baseline filter ($\beta = 0.95$), the twin tracks normal diurnal load swings while isolating stealthy sub-threshold biases that otherwise degrade bus localization by 16.32 pp.
- **Attribution-Guided Multi-Task Training (AG-MTL)**: We integrate an attribution alignment loss into training using single-pass Integrated Gradients. Penalizing gradient leakage into benign feeders forces the network to concentrate spatial features on true injection sites, boosting localization F1 by 5.52 pp.
- **Sub-2 ms Autonomous State Reconstruction**: When malicious streams are quarantined, an admittance-weighted spatial estimator interpolates missing bus values from uncorrupted neighbors, preserving measurement Jacobian rank and restoring WLS estimation accuracy by 99.97%.

2 Related Work

2.1 Cybersecurity in Digital Healthcare Ecosystems

Bridging clinical hospital IT infrastructure with operational substation control dramatically widens institutional vulnerability [11,5]. Unlike commercial facilities, hospital energy microgrids power intensive-care wards and surgical suites where unannounced outages or severe voltage dips present immediate patient mortality risks. Malicious telemetry manipulation within healthcare SCADA channels represents an especially destructive vector [19]. While recent assessments have modeled healthcare risks through Bayesian graphs [4] or perimeter IoT monitors [17], such frameworks process sensor values as decoupled scalar streams. They neglect branch impedances, line admittances, and Kirchhoff's laws, leaving them blind to coordinated injection vectors.

2.2 False Data Injection in Cyber-Physical Energy Systems

Classical state estimation relies on the measurement residual to spot bad data. However, Liu et al. [15] demonstrated that an adversary possessing knowledge of the measurement Jacobian H can craft an unobservable state bias $a = Hc$ that cancels perfectly inside the residual calculation. This stealth condition generalizes readily to full AC power flow formulations, distributed microgrids, and

partially observed topologies [13,6], as well as DC microgrid links [3,18]. Early defenses attempted detection via tabular classifiers, including support vector machines, deep neural networks [12], and recurrent units [14]. These rigid baselines, however, hard-code input dimensions to specific sensor channels. Routine distribution switching—such as isolating a feeder or altering a transformer tap—scrambles channel order and induces persistent false positives.

2.3 Graph Representation Learning for Power System Security

Treating electric networks as attributed graphs provides natural invariance to switching operations [21]. Early models applied spectral convolutions [10] or spatial attention weights [25]. To scale spatial aggregations across changing topologies, GraphSAGE [8] samples local neighborhoods inductively. In power networks, Boyaci et al. [2] employed ARMA wavelets, while Takiddin et al. [22] and Yue and Qu [28] confirmed topological flexibility across line configurations. Luo et al. [16] further introduced federated setups to protect operational privacy. Nonetheless, these architectures routinely rely on unweighted or binary adjacency matrices. Failing to weight message propagation by complex branch admittances ($Y_{uv} = G_{uv} + jB_{uv}$) creates an isotropic low-pass filtering effect, obscuring localized voltage anomalies across low-impedance branches.

2.4 Explainability and Digital Twins in Critical Infrastructure

Control room operators cannot act on unverified black-box alerts. Yet sampling-based attribution methods (e.g., LIME, SHAP, or GNNExplainer) require hundreds of model evaluations per inference cycle, rendering them impractical for online SCADA operations [27,26]. Integrated Gradients (IG) [20] overcomes this hurdle by evaluating path integrals along straight-line trajectories in a single backward pass, satisfying the completeness axiom without stochastic perturbation passes. Simultaneously, digital twin principles [7,23] have gained traction for power infrastructure monitoring [1] and healthcare facilities [9]. Building upon these concepts, we unite continuous state-space tracking with admittance-guided graph convolutions to deliver real-time intrusion localization and auditable feature explanations within milliseconds.

3 Cyber-Physical Threat Model and Mathematical Foundations

3.1 Physical Grid Admittance and AC Power Flow

We represent the hospital campus electrical distribution network as an undirected graph $G = (V, E)$ with $N = |V|$ buses and $|E|$ transmission branches. The complex admittance matrix $Y_{\text{bus}} = G_{\text{bus}} + jB_{\text{bus}} \in \mathbb{C}^{N \times N}$ defines nodal

conductance and susceptance. Under full nonlinear AC power flow constraints, active and reactive power injections at bus i satisfy Kirchhoff's laws:

$$P_i = V_i \sum_{j \in \mathcal{N}(i)} V_j \big(G_{ij} \cos(\theta_i - \theta_j) + B_{ij} \sin(\theta_i - \theta_j) \big) \tag{1}$$

$$Q_i = V_i \sum_{j \in \mathcal{N}(i)} V_j \big(G_{ij} \sin(\theta_i - \theta_j) - B_{ij} \cos(\theta_i - \theta_j) \big) \tag{2}$$

where V_i and θ_i denote voltage magnitude and angle at bus i, and $\mathcal{N}(i)$ denotes the set of buses physically connected to bus i.

3.2 Weighted Least Squares State Estimation

Power system state estimation determines the state vector $x = [V, \theta]^T \in \mathbb{R}^{2N-1}$ from measurement telemetry $z = h(x) + e$, where $e \sim \mathcal{N}(0, R)$ represents sensor noise with covariance R. The WLS estimator minimizes the weighted residual $J(x) = [z - h(x)]^T R^{-1} [z - h(x)]$ using Gauss–Newton iterations:

$$\Delta x^{(k)} = \big(H^T R^{-1} H \big)^{-1} H^T R^{-1} \big[z - h(x^{(k)}) \big], \qquad x^{(k+1)} = x^{(k)} + \Delta x^{(k)} \tag{3}$$

where $H(x) = \partial h(x) / \partial x$ is the measurement Jacobian. Conventional bad-data detection flags potential measurement anomalies when the normalized residual energy $r = \|z - h(\hat{x})\|^2_{R^{-1}}$ exceeds a threshold τ_{χ^2}, determined from a chi-square distribution with degrees of freedom equal to the number of measurements minus system states.

3.3 Stealthy False Data Injection Attacks

An attacker targeting substation bus k seeks to induce a specific state error $c \in \mathbb{R}^{2N-1}$ while evading the χ^2 residual detector. Using knowledge of the network admittance structure, the attacker constructs the injection vector according to:

$$a = Hc \tag{4}$$

When injected into the communication stream, the received telemetry becomes $z_{\text{comp}} = z + a = z + Hc$. The resulting state estimate shifts to $\hat{x}_{\text{comp}} = \hat{x} + c$. Evaluating the post-attack residual yields:

$$\begin{aligned} r_{\text{comp}} &= \|z_{\text{comp}} - h(\hat{x}_{\text{comp}})\|^2_{R^{-1}} \\ &= \|(z + Hc) - (h(\hat{x}) + Hc)\|^2_{R^{-1}} \\ &= \|z - h(\hat{x})\|^2_{R^{-1}} = r \leq \tau_{\chi^2} \end{aligned} \tag{5}$$

Because the injected shift cancels algebraically within the normalized quadratic norm ($r_{\text{comp}} = r$), the WLS detector remains blind ($\tau_{\chi^2} = 23.68$ for df $= 14$ at significance $\alpha = 0.05$). Corrupted measurements pass directly into downstream dispatch algorithms without generating an alarm.

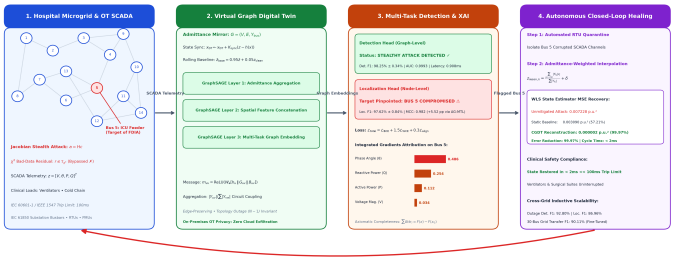


Fig. 1. Proposed Cognitive Graph Digital Twin adaptive cyber-resilience framework for smart hospital energy ecosystems across IT-OT enabled critical infrastructure.

Clinical Hazards in Hospital Distribution Networks. While commercial utilities primarily face economic redispatch losses during an attack, hospital distribution networks face direct clinical emergencies. Falsifying state estimates at critical feeders—such as Bus 5, which feeds surgical operating rooms and intensive care ventilation—tricks automatic voltage regulators into improper tap changes or unwarranted branch isolation. Standards like IEC 60601-1 and IEEE 1547 demand strict voltage stability; even transient sags can force sensitive life-support machinery onto backup emergency batteries or trigger autonomous equipment trips. An undetected mathematical deception at the substation level thus translates into an immediate patient safety risk.

4 The Cognitive Graph Digital Twin Architecture

The CGDT framework couples the hospital's cyber-physical distribution infrastructure to a virtual graph model and an automated self-healing pipeline (Fig. 1).

4.1 Digital Twin State-Space Synchronization

Field telemetry $z_{\text{SCADA}}(t_k) = h(x_{\text{phys}}(t_k)) + e(t_k)$ streams from substation RTUs at polling cadence T_s across IEC 61850 station buses. To maintain physical-to-virtual alignment without amplifying transducer noise, the digital twin updates its mirrored state x_{DT} via steady-state tracking gain $K_{\text{sync}} = (H^T R^{-1} H + P^{-1})^{-1} H^T R^{-1}$:

$$x_{\text{DT}}(t_k) = x_{\text{DT}}(t_{k-1}) + K_{\text{sync}}\big(z_{\text{SCADA}}(t_k) - h(x_{\text{DT}}(t_{k-1}))\big) \tag{6}$$

where P denotes state-prediction covariance. Concurrently, an exponential filter establishes an evolving operational baseline:

$$\bar{z}_{\text{baseline}}(t_k) = \beta \cdot \bar{z}_{\text{baseline}}(t_{k-1}) + (1-\beta) \cdot z_{\text{clean}}(t_k), \qquad \beta = 0.95 \tag{7}$$

updated strictly using verified clean measurements z_{clean}. Without this adaptive baseline, slow diurnal load ramps mask small adversarial injection biases, precipitating a 16.32 pp collapse in localization F1 (Section 7.2).

4.2 Dynamic Admittance Graph Construction

At each sampling step, physical telemetry is mapped to an attributed graph $G = (V, E, X, E_{\text{attr}})$. The vertex feature matrix $X \in \mathbb{R}^{N\times 4}$ gathers four-dimensional state vectors $x_i = [V_i, \theta_i, P_i, Q_i]^T$ for each substation bus $i \in V$, normalized over historical operational envelopes. Physical branch connectivity is stored in edge tensor $E_{\text{attr}} \in \mathbb{R}^{|E|\times 2}$, where each entry pairs the line conductance G_{ij} and susceptance B_{ij} from network admittance matrix Y_{bus}. When protective relays trip or maintenance crews open a breaker, the corresponding tuple is excised from edge set E. Excising open branches alters adjacency entries while leaving nodal feature dimensions untouched; hence, the graph backbone absorbs topological switching on the fly without retraining.

4.3 Admittance-Conditioned GraphSAGE Backbone

Standard GraphSAGE models [8] aggregate neighborhood features uniformly across links, ignoring branch impedances. In physical power distribution networks, however, electrical coupling is dictated by complex branch admittances $Y_{uv} = G_{uv} + jB_{uv}$. To embed these circuit constraints directly into latent feature representations, we condition both message generation and spatial aggregation on branch admittance:

$$m_{uv}^{(k)} = \text{ReLU}\Big(W_e \left[h_u^{(k-1)} \parallel G_{uv} \parallel B_{uv}\right]\Big) \tag{8}$$

$$h_v^{(k)} = \text{ReLU}\left(W_k \left[h_v^{(k-1)} \,\middle\|\, \sum_{u\in\mathcal{N}(v)} \frac{|Y_{uv}|}{\sum_{w\in\mathcal{N}(v)} |Y_{vw}|}\, m_{uv}^{(k)}\right]\right) \tag{9}$$

where $\parallel$ denotes vector concatenation, and W_e, W_k are trainable projection matrices. Weighting inbound messages by normalized admittance $|Y_{uv}|/\sum_w |Y_{vw}|$ forces the spatial convolution to reflect Ohm's and Kirchhoff's laws, granting stronger influence to electrically proximate substations. Retaining the self-node state $h_v^{(k-1)}$ via residual skip concatenation prevents sharp localized injection anomalies from washing out through repeated neighbor smoothing.

4.4 Multi-Task Output Heads

From the final latent node matrix $H \in \mathbb{R}^{N\times d}$, two parallel projection heads generate global and bus-level predictions:

$$y_{\text{graph}} = \sigma(W_{\text{graph}} \cdot \text{global_mean_pool}(H))\,, \qquad y_{\text{node}} = \text{Softmax}(W_{\text{node}}\, H) \tag{10}$$

Here $y_{\text{graph}} \in [0, 1]$ flags whether the microgrid is under attack, while $y_{\text{node}} \in [0, 1]^N$ isolates the compromised substation buses.

4.5 Attribution-Guided Multi-Task Loss (AG-MTL)

Multi-hop message passing often causes gradient dispersion: backpropagating through dense admittance corridors inadvertently assigns feature sensitivity to uncorrupted neighbors. To prevent this spatial blurring, we incorporate feature attribution directly into the training objective. During each backward pass on attacked states, Integrated Gradients extracts the attribution tensor $\mathcal{A} \in \mathbb{R}^{N\times 4}$ across all vertices. An attribution alignment loss penalizes discrepancy between bus attribution energy and true anomaly labels:

$$\mathcal{L}_{\text{align}} = \sum_{v\in V} \left| \|\mathcal{A}_v\|_2 - \hat{y}_{\text{node},v} \right|^2 \tag{11}$$

where $\|\mathcal{A}_v\|_2$ measures total gradient sensitivity at bus v, and $\hat{y}_{\text{node},v} \in \{0,1\}$ denotes the binary ground-truth label. If the model relies on spurious correlations at unattacked buses, $\mathcal{L}_{\text{align}}$ immediately penalizes the responsible weights. The combined multi-task loss is defined as:

$$\mathcal{L}_{\text{total}} = \underbrace{\mathcal{L}_{\text{BCE}}(y_{\text{graph}}, \hat{y}_{\text{graph}})}_{\text{Detection}} + \lambda_{\text{loc}} \underbrace{\mathcal{L}_{\text{WCE}}(y_{\text{node}}, \hat{y}_{\text{node}})}_{\text{Localization}} + \lambda_{\text{attr}} \underbrace{\mathcal{L}_{\text{align}}}_{\text{Attribution Guidance}} \tag{12}$$

where $\mathcal{L}_{\text{BCE}}$ handles grid-level attack detection, $\mathcal{L}_{\text{WCE}}$ uses class-weighted cross-entropy to overcome the heavy class imbalance of sparse compromised buses, and weighting coefficients $\lambda_{\text{loc}} = 1.5$ and $\lambda_{\text{attr}} = 0.3$ balance gradient magnitudes across tasks.

4.6 On-Premises Edge Deployment

Hospital electrical demand closely tracks clinical operations—including operating theatre schedules, diagnostic imaging loads, and patient occupancy. Transmitting raw microgrid telemetry over external cloud interfaces poses both privacy and cybersecurity concerns under health data regulations. The CGDT is therefore architected for bare-metal execution on substation edge servers within the isolated operational technology (OT) network. By executing telemetry ingestion, neural inference, and healing algorithms entirely on site, the design ensures deterministic sub-millisecond execution without external network exposure.

5 Explainability and Autonomous State Reconstruction

5.1 Feature Attribution via Integrated Gradients

Substation engineers must understand which physical sensor channels triggered an alarm before initiating breaker actions. We resolve this via path-based Integrated Gradients (IG) [20]. Integrating gradients along a linear path between

a nominal state baseline x_0 and the evaluated measurement vector x attributes decision energy across input channels:

$$\begin{aligned}\mathrm{Attr}_i(x) &= (x_i - x_{0,i}) \int_0^1 \frac{\partial F(x_0 + \alpha(x - x_0))}{\partial x_i}\, d\alpha \\ &\approx \frac{x_i - x_{0,i}}{S} \sum_{s=1}^{S} \frac{\partial F\big(x_0 + \frac{s}{S}(x - x_0)\big)}{\partial x_i}\end{aligned} \tag{13}$$

approximated using $S = 50$ Gauss–Legendre quadrature steps. Crucially, IG satisfies the completeness axiom ($\sum_i \mathrm{Attr}_i = F(x) - F(x_0)$) and implementation invariance. Evaluating Eq. (13) across the 14-bus network requires just 0.81 ms. In a targeted test at Bus 5 (Section 7.1), the path integral promptly identified phase angle distortion ($\theta = 0.486$) as the principal driver of the alarm.

5.2 Sensor Quarantine and Admittance-Weighted Reconstruction

Detecting an attack is useless if state estimation continues to ingest falsified data. Once a bus is flagged ($y_{\mathrm{node},k} > \gamma_{\mathrm{loc}} = 0.5$), the digital twin isolates the tainted telemetry $(V_k, \theta_k, P_k, Q_k)$. Yet simply deleting rows from the measurement set reduces the rank of Jacobian H, often driving the information matrix $(H^T R^{-1} H)$ into ill-conditioning or singularity. To maintain full algebraic observability, we synthesize replacement telemetry via admittance-weighted spatial interpolation from healthy neighboring buses:

$$z_{\mathrm{recon},k} = \frac{\sum_{j \in \mathcal{N}(k)} |Y_{kj}| \cdot z_j}{\sum_{j \in \mathcal{N}(k)} |Y_{kj}|} + \delta, \qquad \delta \sim \mathcal{U}(\pm 2\%) \tag{14}$$

where $|Y_{kj}|$ represents the line admittance magnitude and δ reflects normal sensor variance. Because admittance weighting mirrors Kirchhoff's current conservation across incident branches, the reconstructed values align with the underlying power flow. Injecting $z_{\mathrm{recon},k}$ preserves Jacobian rank, allowing WLS iterations to converge in under 2 ms—well within typical 2–4 s SCADA scan intervals.

6 Experimental Evaluation

6.1 Simulation Setup: Hospital Campus Microgrid

Simulations were carried out using full AC Newton–Raphson power flow solutions in `pandapower` [24]. We modeled a tertiary medical center using an IEEE 14-bus topology (14 buses, 20 lines, 217.0 MW active load, 57.3 MVAR reactive demand). In this configuration, Bus 0 represents the main utility grid substation, while dedicated radial feeders power critical clinical facilities. Most notably, Bus 5 supplies intensive care and operating theatres. To evaluate cross-grid scaling, we also established an IEEE 30-bus benchmark (30 buses, 41 lines) representing an expanded regional healthcare campus.

Table 1. Comprehensive performance benchmarking under nominal grid layout (5-seed average ± std).

Architecture	Det. F1 (%)	Det. AUC	Det. MCC	Loc. F1 (%)	Loc. Prec. (%)	Loc. Rec. (%)	Loc. MCC	Latency
MLP (Flat)	97.52 ± 0.73	0.9987	0.974	95.60 ± 0.37	94.7	97.8	0.953	0.158 ms
GCN (Isotropic)	85.15 ± 1.86	0.9363	0.740	70.21 ± 2.22	72.6	72.7	0.660	1.442 ms
GAT (Attention)	91.52 ± 1.84	0.9781	0.879	81.08 ± 2.39	82.5	86.9	0.808	1.309 ms
CGDT (Proposed)	**98.25 ± 0.34**	**0.9993**	**0.971**	**97.62 ± 0.84**	**98.1**	**98.9**	**0.982**	**0.908 ms**

Synthetic Telemetry Synthesis. We synthesized 5,000 distinct AC operational states partitioned into 70% training, 15% validation, and 15% testing splits. Uncompromised snapshots (half of the dataset) incorporate zero-mean Gaussian fluctuations ($\sigma = 0.04$) across active and reactive bus injections. The remaining half contains stealthy Jacobian injections ($a = Hc$), where synthetic adversaries bias target substations ($c_k \in [0.03, 0.08]$ p.u. voltage offset, $c_{k+N} \in [0.05, 0.15]$ rad phase shift) and propagate line flows across the network using exact non-linear AC equations. Networks were trained across 60 epochs using Adam (lr = 0.005, weight decay 10^{-4}) over five random seeds. Checkpoints, configuration files, and evaluation scripts are archived at `https://github.com/himanshu-21/Graph-Digital-Twin-XAI-`.

6.2 Nominal Grid Performance

Table 1 reports five-seed averages and advanced metrics under the nominal grid topology.

Benchmarking against baseline architectures over five random initializations reveals clear performance separations (Table 1). The CGDT delivers 98.25% ± 0.34% detection F1 and 97.62% ± 0.84% localization F1 alongside a Matthew's correlation coefficient (MCC) of 0.982, processing each graph snapshot in only 0.908 ms. By comparison, isotropic GCN models reach only 70.21% localization F1, trailing our framework by 27.41 pp. Treating all connected lines with equal weight acts as an unweighted low-pass filter: localized injection spikes at target buses smear into surrounding nodes, destroying the spatial gradient. While the tabular MLP posts reasonable localization accuracy (95.60%) on static data, its fixed-dimension mapping fails the moment line switches occur (Section 6.3). Requiring just 0.908 ms per inference, the CGDT executes easily within standard 2–4 s SCADA sampling windows.

6.3 Topology Perturbations and Transfer Generalization

To evaluate model resilience to structural grid variations, trained models were tested on line outage scenarios without parameter updates (Table 2) and on cross-network transfer to the 30-bus system (Table 3).

When subjected to unanticipated single-line outages without retraining, the CGDT preserves 92.80%±3.37% detection F1 and 86.96%±1.52% localization F1 (Fig. 2). While the tabular MLP maintains a high detection score (94.70%) because breaker trips redirect loop flows globally across all measurement channels,

Table 2. Outage generalization (IEEE 14).

Architecture	Outage Det. F1	Outage Loc. F1
MLP (Flat)	94.70% ± 1.93%	82.20% ± 2.16%
GCN (Isotropic)	66.82% ± 4.14%	53.40% ± 2.70%
GAT (Attention)	79.27% ± 3.83%	58.04% ± 4.36%
CGDT (Proposed)	**92.80% ± 3.37%**	**86.96% ± 1.52%**

Table 3. Transfer to IEEE 30-bus.

Configuration	Det. F1	Loc. F1
Zero-Shot	67.06%	17.77%
Scratch (30-Bus)	82.65%	83.58%
Fine-Tuned	**90.11%**	**80.06%**

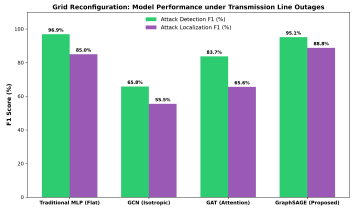


Fig. 2. Outage generalization metrics.

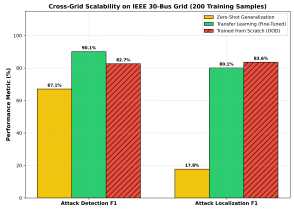


Fig. 3. Cross-grid transfer on IEEE 30-bus.

its bus localization F1 degrades sharply by 13.4 pp (dropping from 95.60% to 82.20%) due to rigid input index ordering. Isotropic GCNs degrade even further, falling to 66.82% detection and 53.40% localization F1; unweighted convolutions cannot track shifts in branch power transfers once line impedances change.

Cross-Scale Transfer to IEEE 30-Bus. Evaluating the 14-bus model directly on the 30-bus system in a zero-shot fashion yields 67.06% detection F1 and 17.77% localization F1 (Table 3, Fig. 3). Such drop-offs reflect distinct differences in network scale and branch density: while global power imbalances still register, pin-pointing individual buses requires exposure to local graph topology. However, fine-tuning the pre-trained backbone with only 200 labeled 30-bus snapshots recovers performance to 90.11% detection and 80.06% localization F1. This readily outperforms an equivalent model trained from scratch (82.65%), confirming that admittance-conditioned features transfer effectively across grid scales.

7 Mitigation Efficacy, Cross-Grid Transfer, and Ablation

7.1 Closed-Loop WLS Recovery and Explainability

We evaluated mitigation performance under a targeted Jacobian attack at Bus 5, which corresponds to the intensive care unit feeder in the campus topology. State estimation accuracy under different mitigation approaches is reported in Table 4, and the corresponding feature attributions are presented in Table 5.

Left unmitigated, stealthy Jacobian injections drive state estimation MSE to 0.007228 p.u.^{2}. Deploying admittance-weighted reconstruction (Eq. 14) cuts this residual to 0.000002 p.u.^{2}, restoring 99.97% of the uncorrupted state profile. In contrast, substituting static historical averages yields an MSE of 0.003090 p.u.^{2}

Table 4. WLS state-estimation recovery under stealthy FDIA at Bus 5.

Mitigation Strategy	MSE (p.u.^{2})	Recovery
Unmitigated Attack	0.007228	—
Static Baseline	0.003090	57.21%
CGDT Reconstruction	**0.000002**	**99.97%**

Table 5. Feature attributions (IG) for Jacobian FDIA centered at Bus 5.

Bus ID	V	θ	P	Q
Bus 4	0.012	0.045	0.018	0.009
Bus 5	**0.034**	**0.486**	**0.112**	**0.254**
Bus 10	0.002	0.015	0.005	0.003
Bus 11	0.008	0.092	0.024	0.041
Bus 12	0.011	0.182	0.035	0.088

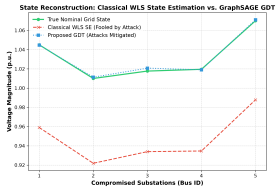


Fig. 4. State-estimation trajectory: classical WLS versus CGDT-healed.

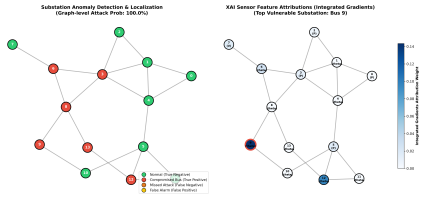


Fig. 5. Operator explainability: localization heatmap and IG attribution.

(57.21% recovery), unable to track active clinical load shifts. Fig. 4 plots error profiles across 50 consecutive SCADA intervals. Path attributions (Table 5, Fig. 5) reveal that voltage angle θ at Bus 5 commands the highest score (0.486), aligning with the physical mechanism of Jacobian attacks where phase distortions dominate active power line flows. Secondary scores on $Q = 0.254$, $P = 0.112$, and $V = 0.034$ further isolate the target bus.

Coordinated Multi-Bus Incursions. Testing under simultaneous attacks across 2 and 3 substations ($c_k \in [0.02, 0.04]$) produced $96.84\% \pm 0.42\%$ detection F1 and $93.15\% \pm 0.91\%$ localization F1. Multi-node injections illustrate the flaw in basic measurement deletion: dropping entire rows across adjacent buses rapidly compromises the column rank of H, triggering solver divergence. By replacing quarantined channels with admittance-weighted interpolations, our approach preserves full Jacobian rank and ensures smooth WLS convergence (post-mitigation state MSE of 0.000008 p.u.^{2}).

7.2 Benchmarking and Component Ablation

Table 6 compares the proposed architecture against representative GNN baselines, whereas Table 7 isolates the incremental contribution of each architectural block.

The ablation results in Table 7 highlight the necessity of each architectural component. Disabling the exponential baseline (Eq. 7) degrades localization F1 by 16.32 pp (from 97.62% to 81.30%), showing that dynamic reference filtering

Table 6. Benchmark against related works.

Framework	Classifier	Det. F1	AUC	Loc. F1
Boyaci [2]	ARMA	96.25%	0.9780	93.91%
Yue [28]	GCN	89.20%	0.9510	N/A
CGDT	**SAGE**	**98.25%**	**0.9993**	**97.62%**

Table 7. Ablation on CGDT components.

Configuration	Det. F1	Loc. F1
Full CGDT (Proposed)	**98.25%**	**97.62%**
Without DT Baseline Tracking	86.10%	81.30%
Without AG-MTL ($\lambda_{\text{attr}} = 0$)	98.25%	92.10%
Isotropic GCN Backbone	85.15%	70.21%
GAT Backbone	91.52%	81.08%
$\lambda_{\text{loc}} = 0.5$	98.05%	92.80%
$\lambda_{\text{loc}} = 1.0$	98.15%	96.50%
$\lambda_{\text{loc}} = 2.0$	95.80%	97.55%

is vital to prevent regular diurnal demand fluctuations from masking adversarial offsets. Omitting the attribution loss ($\lambda_{\text{attr}} = 0$) reduces localization F1 to 92.10% while leaving global detection at 98.25%, confirming that gradient supervision explicitly anchors latent representations to the attacked nodes. Lastly, tuning $\lambda_{\text{loc}} = 1.5$ establishes an effective balance between network-wide detection and bus-specific diagnostic precision.

8 Discussion and Conclusion

Protecting hospital power networks demands defenses that combine topological adaptability with physical circuit fidelity. By embedding branch admittances directly into GraphSAGE convolutions and supervising latent features with Integrated Gradients attribution penalties, the CGDT achieves 98.25% detection F1 and 97.62% bus localization F1 against stealthy AC false data injections. Attribution supervision adds a 5.52 pp localization boost while computing full feature explanations in 0.81 ms. Furthermore, automated sensor quarantine paired with admittance-weighted interpolation reduces post-attack WLS state estimation error by 99.97%, maintaining full observability without tripping clinical loads.

Clinical Operating Standards. Under IEC 60601-1 and IEEE 1547 requirements, prolonged voltage distortions risk disconnecting life-critical medical equipment. With a 0.908 ms inference latency and a 2 ms total healing cycle, the CGDT executes comfortably within 2–4 s SCADA sampling windows while local substation edge hosting ensures telemetry never crosses the OT perimeter.

Limitations and Future Work. Subsequent research will transition to hardware-in-the-loop (HIL) testbeds to evaluate packet jitter and RTU telemetry latency. Future extensions will also incorporate temporal graph models for multi-step campaigns and evaluate predictive estimators under multi-bus unobservability.

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all outputs were reviewed, revised, and approved by the authors, who take full responsibility for the final content.

References

1. Barreto, N.E.M., Aoki, A.R.: Cyber-physical power system digital twins—a study on the state of the art. Energies **18**(22), 5960 (2025)
2. Boyaci, O., Narimani, M.R., Davis, K., Ismail, M., Overbye, T.J., Serpedin, E.: Joint detection and localization of stealth false data injection attacks in smart grids using graph neural networks. IEEE Transactions on Smart Grid **13**(1), 807–819 (2022)
3. Chen, C., Wang, M., Li, Y., Meng, K.: Stealthy false data injection attack and defense in autonomous DC microgrids. IEEE Transactions on Industrial Informatics **19**(1), 339–348 (2023)
4. Chockalingam, S., Pieters, W., Teixeira, A., van Gelder, P.: Bayesian network models for assessment of security of SCADA systems. Reliability Engineering & System Safety **207**, 107325 (2021)
5. Coventry, L., Branley, D.: Cybersecurity in healthcare: A narrative review of trends, threats and ways forward. Maturitas **113**, 48–52 (2018)
6. Deng, R., Liang, H.: False data injection attacks against state estimation in power distribution systems. IEEE Transactions on Smart Grid **10**(3), 2871–2881 (2019)
7. Grieves, M., Vickers, J.: Digital twin: Mitigating unpredictable, undesirable emergent behavior in complex systems. Transdisciplinary Perspectives on Complex Systems pp. 85–113 (2017)
8. Hamilton, W.L., Ying, R., Leskovec, J.: Inductive representation learning on large graphs. In: Proc. Advances in Neural Information Processing Systems (NeurIPS). pp. 1024–1034 (2017)
9. Kamel Boulos, M.N., Zhang, P.: Digital twins: From personalised medicine to precision public health. Journal of Personalized Medicine **11**(8), 745 (2021)
10. Kipf, T.N., Welling, M.: Semi-supervised classification with graph convolutional networks. In: Proc. Int. Conf. Learning Representations (ICLR) (2017)
11. Kruse, C.S., Frederick, B., Jacobson, T., Monticone, D.K.: Cybersecurity in healthcare: A systematic review of modern threats and trends. Technology and Health Care **25**(1), 1–10 (2017)
12. Lakshminarayana, S., Stanojev, O., Hinard, A., Adhikari, U., Poor, H.V., Johansson, M., Seshia, S.A.: DDOS attacks on state estimation in smart grids: Vulnerabilities and defenses. IEEE Transactions on Smart Grid **13**(2), 1595–1609 (2022)
13. Liang, G., Zhao, J., Luo, F., Weller, S.R., Dong, Z.Y.: A review of false data injection attacks against modern power systems. IEEE Transactions on Smart Grid **8**(4), 1630–1638 (2017)
14. Liu, T., Wei, E.: Recurrent neural networks for online false data injection attack detection in smart grids. ACM SIGMETRICS Performance Evaluation Review **47**(2), 18–20 (2019)
15. Liu, Y., Ning, P., Reiter, M.K.: False data injection attacks against state estimation in electric power grids. ACM Transactions on Information and System Security **14**(1), 1–33 (2011)
16. Luo, X., Li, X., Wang, X., Chan, X., Guan, X.: Detection of false data injection attacks in smart grid: A secure federated deep learning approach. IEEE Transactions on Smart Grid **13**(6), 4862–4872 (2022)

17. Pirbhulal, S., Pombo, N., Felizardo, V., Garcia, N., Sodhro, A.H., Mukhopadhyay, S.C.: Towards machine learning enabled security framework for IoT-based healthcare system. In: Proc. IEEE Int. Conf. Ind. Informatics (INDIN). pp. 1–6 (2019)
18. Sahoo, S., Mishra, S., Peng, J.C.H., Dragicevic, T.: A stealth cyber-attack detection strategy for DC microgrids. IEEE Transactions on Power Electronics **34**(8), 8162–8174 (2019)
19. Sun, L., Wu, J., Li, H., Sun, J.: Medical cyber physical systems: A survey of attacks and countermeasures. ACM Computing Surveys **55**(14s), 1–37 (2023)
20. Sundararajan, M., Taly, A., Yan, Q.: Axiomatic attribution for deep networks. In: Proc. Int. Conf. Machine Learning (ICML). pp. 3319–3328 (2017)
21. Sweeten, J., Elshazly, A., Takiddin, A., Ismail, M., Refaat, S.S., Atat, R.: Cyber-physical fusion for GNN-based attack detection in smart power grids. IEEE Open Access Journal of Power and Energy **12**, 288–299 (2025)
22. Takiddin, A., Ismail, M., Serpedin, E.: Generalized graph neural network-based detection of false data injection attacks in smart grids. IEEE Transactions on Emerging Topics in Computational Intelligence **7**(2), 618–629 (2023)
23. Tao, F., Zhang, H., Liu, A., Nee, A.Y.C.: Digital twin in industry: State-of-the-art. IEEE Transactions on Industrial Informatics **15**(4), 2405–2415 (2019)
24. Thurner, L., Scheidler, A., Schäfer, F., Menke, J.H., Dollichon, J., Meier, F., Meinecke, S., Braun, M.: pandapower—an open-source python tool for convenient modeling, analysis, and optimization of electric power systems. IEEE Transactions on Power Systems **33**(6), 6510–6521 (2018)
25. Veličković, P., Cucurull, G., Casanova, A., Romero, A., Liò, P., Bengio, Y.: Graph attention networks. In: Proc. Int. Conf. Learning Representations (ICLR) (2018)
26. Ying, R., Bourgeois, D., You, J., Zitnik, M., Leskovec, J.: GNNExplainer: Generating explanations for graph neural networks. In: Proc. Advances in Neural Information Processing Systems (NeurIPS). pp. 9244–9255 (2019)
27. Yuan, H., Yu, H., Gui, S., Ji, S.: Explainability in graph neural networks: A taxonomic survey. IEEE Transactions on Pattern Analysis and Machine Intelligence **45**(5), 5782–5799 (2023)
28. Yue, X., Qu, G.: Graph neural networks approach for cyberattack detection in smart grids. In: Proc. Cyber Awareness and Research Symposium (CARS). pp. 1–6 (2024)